

FRITZLAR, Germany

WMO: 104390

Lat: 51.1189N

Lon: 9.2800E

Elev: 595

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.7	17.7	8.7	8.8	15.3	13.3	11.1	19.2	23.8	43.6	21.1	44.0	3.8	310	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	86.9	68.0	82.7	66.2	79.1	64.6	69.6	82.5	67.8	79.6	66.2	76.8	7.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	95.4	73.8	63.9	90.9	72.4	62.3	85.8	70.5	33.9	82.5	32.5	79.6	31.2	76.8	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.7	16.3	14.2	DB	7.8	92.7	6.7	3.3	3.0	95.1	-0.9	97.0	-4.7	98.8	-9.6	101.2
			WB	7.4	73.0	6.5	1.9	2.7	74.4	-1.0	75.5	-4.7	76.6	-9.4	78.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	33.7	35.6	41.5	48.7	55.6	61.3	65.2	64.7	57.5	49.7	41.6	35.5	(6)
	DBStd	12.96	8.47	7.78	6.61	6.83	6.40	5.93	5.84	5.58	5.42	6.34	6.40	7.72	(7)
	HDD50	2107	504	403	271	106	20	1	0	0	5	85	261	451	(8)
	HDD65	5924	969	822	729	491	297	142	73	75	233	475	703	915	(9)
	CDD50	1847	0	1	7	66	195	339	470	455	230	75	8	1	(10)
	CDD65	189	0	0	0	0	7	31	78	65	8	0	0	0	(11)
	CDH74	2271	0	0	0	16	119	395	877	727	134	3	0	0	(12)
	CDH80	686	0	0	0	1	17	98	298	245	27	0	0	0	(13)
(14) Wind	WSAvg	6.1	7.0	6.7	6.9	6.3	5.8	5.7	5.7	5.5	5.4	5.7	6.0	6.7	(14)
(15) Precipitation	PrecAvg	26.4	2.0	1.7	1.7	1.7	2.5	2.6	3.0	2.4	2.2	2.1	2.3	2.2	(15)
	PrecMax	35.2	4.3	4.3	3.1	3.1	5.9	4.5	5.2	5.4	4.9	6.0	3.5	4.7	(16)
	PrecMin	19.3	0.2	0.5	0.2	0.1	0.7	1.1	0.9	0.7	0.2	0.4	0.1	0.6	(17)
	PrecStd	3.7	1.0	0.9	0.8	0.7	1.3	0.8	1.1	1.2	1.1	1.2	0.9	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	55.6	65.5	76.4	82.2	87.8	92.0	92.0	84.1	71.5	59.4	54.6	(19)
		MCWB	49.4	48.1	52.1	58.6	65.3	69.0	68.7	68.2	66.0	61.6	54.4	50.1	(20)
	2%	DB	50.3	52.0	60.5	71.3	77.2	82.6	87.6	86.3	78.2	65.7	56.3	51.6	(21)
		MCWB	46.6	46.8	50.0	56.8	62.5	67.4	68.1	67.1	63.5	59.1	52.1	48.4	(22)
	5%	DB	47.8	49.7	55.8	66.6	73.3	78.6	82.7	81.9	72.7	62.3	53.6	48.5	(23)
		MCWB	44.9	45.3	48.2	54.3	59.6	64.9	66.4	65.6	61.6	57.3	50.3	46.1	(24)
	10%	DB	44.9	46.8	52.2	62.4	69.5	74.8	78.8	77.2	68.2	59.4	50.9	46.2	(25)
		MCWB	42.4	43.2	46.5	52.6	58.2	62.9	64.7	63.9	59.4	55.3	48.0	44.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	49.7	54.9	60.6	68.2	72.0	71.9	70.6	67.5	62.3	55.5	51.3	(27)
		MCDB	53.1	53.0	61.7	71.6	79.5	83.5	85.2	85.3	79.1	68.6	58.2	53.8	(28)
	2%	WB	47.2	47.6	51.5	57.9	64.2	68.8	69.6	68.7	64.8	59.9	52.8	48.7	(29)
		MCDB	49.7	51.5	57.6	68.7	74.1	79.5	82.6	82.2	74.9	64.8	55.9	51.1	(30)
	5%	WB	44.7	45.5	49.3	55.5	61.6	66.4	67.8	67.0	62.5	57.7	50.4	46.1	(31)
		MCDB	47.3	49.1	55.2	64.6	70.8	76.4	79.8	78.6	70.8	61.6	53.1	48.5	(32)
	10%	WB	42.3	43.3	46.9	53.2	59.1	64.1	66.0	65.2	60.6	55.8	48.3	43.8	(33)
		MCDB	44.7	46.7	52.1	61.1	66.8	72.8	76.5	74.9	67.1	59.3	50.7	45.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	12.1	15.4	19.5	19.8	20.1	20.6	20.8	18.4	14.7	9.8	8.6	(35)
		MCDBR	9.7	16.0	23.6	27.8	27.7	28.1	28.9	29.5	27.0	18.8	12.5	10.0	(36)
	5% WB	MCWBR	8.1	11.4	14.7	15.1	14.4	14.2	12.2	12.7	14.0	12.5	9.7	8.4	(37)
		MCWBR	9.6	14.1	20.9	24.4	24.6	25.6	25.4	24.9	22.7	17.7	11.8	9.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.309	0.333	0.368	0.406	0.411	0.409	0.422	0.415	0.382	0.354	0.333	0.306	(40)
		taud	2.411	2.368	2.300	2.229	2.250	2.272	2.241	2.282	2.360	2.425	2.421	2.406	(41)
	Ebn at Noon		227	251	262	264	268	269	263	258	254	240	214	208	(42)
		Edh at Noon	19	26	33	40	41	41	41	38	31	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		227	444	797	1273	1519	1667	1580	1354	974	569	256	169	(44)
	RadStd		33	69	110	158	164	154	206	126	134	60	37	20	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (44 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-214	+86	N/A	

Nomenclature: See separate page